Nguyen Viet Anh

Data Engineer | Applied Machine Learning | Ho Chi Minh City
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Summary

Data Engineer with experience in Oracle ETL, SQL, and Python development. My personal and academic projects include training and evaluating models with LightGBM and PyTorch, comparing OCR methods, and building an automated prediction-scoring tool. I focus on data preparation, validation, and making ML experiments reproducible.

Education

Ho Chi Minh City University of Technology (HCMUT) 2022 – 2026
Vietnam National University Ho Chi Minh City (VNU-HCM).
Computer Science (Information Systems) – Expected graduation: Nov 2026.

Experience

Data Engineer – OSAS Joint Stock Company Jun 2025 – Present

- Develop Oracle ETL from three source systems with PL/SQL and ODI 12c, including incremental loads and SCD Type 2 dimensions; configured GoldenGate CDC for a separate public-sector warehouse.
- Reconcile source and warehouse records and correct policy-counting and sales-channel classification errors.
- Built a Vietnamese financial-statement assistant with Python, FastAPI and Ollama, combining summaries and financial-ratio extraction with SQL/vector retrieval, source-page references and an ingestion queue.

Projects

Adult Income – Classical ML and PyTorch Oct 2026
Coursework and later personal extension; UCI Adult | GitHub

- Checked duplicate feature rows and conflicting labels; fitted preprocessing on training only and selected models and thresholds on validation.
- Evaluated LightGBM and PyTorch models on 16,249 test records: LightGBM achieved 86.73% accuracy and 0.9276 ROC-AUC; MLP accuracy was 84.09–84.50% across three seeds.
- Built a public-validation prediction-scoring tool with schema, ID and probability checks; added 50 automated tests covering scoring, early stopping and PyTorch checkpoint restoration.

HealthMate – Wearable Health Monitoring Sep 2025 – May 2026
Graduation thesis; team of three | Backend | Flutter app

- Designed the Flutter interface and helped evaluate stress models using 869 labeled windows from 15 WESAD participants. Held out one participant at a time to assess performance on people outside the training set.
- Evaluated prescription OCR on 66 manually labeled samples. PaddleOCR with OpenCV and extraction rules achieved medication-name micro-F1 of 0.46, compared with 0.24 for PaddleOCR-VL; used the comparison to guide the app's OCR choice.

Vietnam Stock Data Pipeline 2025 – 2026
Personal project; streaming and warehouse | GitHub

- Connected VNStock, Kafka, PySpark Structured Streaming, TimescaleDB and Snowflake; used incremental sync, persisted watermarks, Bronze/Silver/DW layers and Streams & Tasks.
- Built Streamlit/Plotly dashboards and Docker Compose services with Nginx, ZooKeeper, Kafka/Spark monitoring, pgAdmin and Linux/systemd operations.

Technical Skills

ML & Evaluation: Python, NumPy, Pandas, scikit-learn, LightGBM, PyTorch, CatBoost, imbalanced-learn.
Data Engineering: SQL, PL/SQL, Oracle, ODI, GoldenGate, Kafka, PySpark, PostgreSQL/TimescaleDB, Snowflake.
Applications & Tools: FastAPI, Flutter/Dart, OpenCV, PaddleOCR, Streamlit, Plotly, Git, Linux, Docker Compose.

Certifications

Python Project for Data Engineering – IBM/Coursera, 2026.
Relational Database Administration (DBA) – IBM/Coursera, 2026. | All certificates